

Virginia Tech's Dynamic Anthropomorphic Robot with Intelligence—Open Platform (DARwIn-OP) is another recent example of an open-platform miniature humanoid robot that's been developed for research, education and outreach. Users are encouraged to modify and use both the hardware and software aspects of DARwIn-OP. CAD files for all its parts, and instruction manuals for fabrication and assembly are available for free on the university's website, and various software implementations are possible using C++, Python, LabVIEW, MATLAB, etc. The platform features advanced computational power, sophisticated sensors, a high payload capacity and dynamic motion ability.

All of Willow Garage's projects, including PR2 and Turtlebot, are also typical open source examples. The robot operating system, ROS; the open-source computer vision libraries for real-time perception, called OpenCV; other components published by Stanford University's Artificial Intelligence Laboratory and Steve Cousins' Willow Garage have become quite popular in the robotics field.

Gostai, a French company that makes surveillance and telepresence robots marketed as part of the Jazz brand of products, is another open-source success story. With its tagline 'Robotics for Everyone', the company has not only demonstrated the benefits of using open components, but given back to the community by fully open-sourcing its prized piece of technology, Urbi—an open-source software platform to control robots.

Rohit de Sa points out two other interesting examples, "In the mechanical branch, MakerBot's Thing-O-Matic and RepRap's Mendel are both completely open-source 3D printers. You can actually print out mechanical parts using melted plastic! Incidentally, the Thing-O-Matic and the Mendel, both use other OSH for their control electronics. So what we're seeing here is a symbiotic relation between multiple branches of engineering. Electronics feeds mechanical engineering, which feeds something else, and so on and so forth."

Take your pick

One of the biggest examples of OSH, which is widely used by the robotics community in India, is Arduino -- an open-source electronics prototyping platform based on flexible, easy-to-use hardware and software. The Arduino board can receive inputs from a variety of sensors, and in turn control external factors like lights, motors and actuators. Arduino's reference designs are available on the company's website as CAD files, and can be modified to suit the user's needs. The board itself can either be bought assembled or built by the user based on the designs. The microcontroller on the board is programmed using the Arduino programming language and the Arduino development environment, all of which can be downloaded for free from the Arduino website <http://www.arduino.cc/>.



The RoboMSR brigade (Left to Right) : Praveen Pitchai, Shekar N H, Vineet Sahu

"The Arduino platform is creating a revolution in the field of robotics, as it is simple to use. The board is also flexible, in the sense that there can be different voltages taken from the same board, there can be digital signals and analogue signals given as inputs, the size of the microcontroller can vary from being an Atmega8 to an Atmega2560, and so on. Plus, there are various interfaces for different hardware components, from the very basic to the more complex. There are also various types of boards available for different utilities. There are many projects that are being carried out with the help of this board, for hobbyists as well as for researchers," says Pitchai.

Arduino is quickly gaining traction in India, and growing to be the favourite platform for low-end robotics and learning kits. Azad says his start-up is also using Arduino widely in its projects: "We sell open-source boards based on Arduino, and teach students robotics through it. In a few years' time, Arduino will become a standard platform for teaching microcontrollers and robotics to students in colleges."

Another popularly used platform in India is the Beagle Board, an interesting piece of OSH developed by a skilled community including several employees of Texas Instruments (TI). The Beagle Board is a low-cost, fan-less, single-board computer based on low-power TI processors, featuring the ARM Cortex-A8. It is known for being light but highly expandable. Jayakrishnan says that Energid uses the Beagle Board for remote-control video processing in some of its products.

BITS-Pilani's Centre for Robotics and Intelligent Systems (CRIS) has developed microcontroller hardware for use by students in their courses. "All designs, software and source code are released under the Creative Commons Attribution Share-Alike (India) licence for use by students of BITS, and anyone else," says de Sa.

A lot of open-source software is also used in robotics. We have already mentioned some software platforms such as ROS, OpenCV and Urbi. OpenRAVE is another open-source, cross-platform, plug-in-based robot planning architecture that serves all planning and execution purposes. It includes services like collision detection, robot kinematics, physics, robot controls, and a network scripting environment. This was developed at Carnegie Mellon University.

OpenJAUS and RI-JAUS software development kits are used in the joint architecture for unmanned systems,